



Artificial Intelligence Capabilities and Trust as Determinants of Continuance Intention to Use Mobile Banking

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Abstract: The rapid integration of artificial intelligence (AI) into mobile banking applications has considerably transformed digital financial services, shifting the primary challenge from user adoption to sustaining long-term usage. In emerging digital banking markets such as Indonesia, continuance intention has become critical to the development of mobile banking. The purpose of this study is to examine, from a post-adoption perspective, the effects of artificial intelligence capabilities and trust on continuance intention in mobile banking. A quantitative research design was employed to conduct a cross-sectional survey of 150 mobile banking users in Indonesia. The results obtained from Partial Least Squares Structural Equation Modeling (PLS-SEM) showed that both artificial intelligence capabilities and trust had significant positive effects on continuance intention in mobile banking. More specifically, users' perceptions of artificial intelligence capabilities, such as personalization, responsiveness, automation, and learning ability, all played a crucial role in reinforcing continued usage. In addition, trust, as a core psychological determinant, directly affected users' willingness to rely on AI-enabled mobile banking and to be loyal to such services. Simply put, technological competence alone was not sufficient to sustain long-term usage without corresponding levels of user trust. Therefore, the development of advanced AI functionality and trust-building strategies should be aligned. This study contributes to the literature on mobile banking and information systems by conducting post-adoption research through the integration of artificial intelligence capabilities and trust within a parsimonious research model. With a focus on continuance intention rather than initial adoption, the study provided a more relevant explanation for user behavior in a competitive digital banking environment. The findings offered convincing and practical insights for banks and fintech providers to ensure long-term sustainability of mobile banking services.

Keywords: Artificial intelligence capabilities; Trust; Continuance intention; Mobile banking

1 Introduction

The widespread diffusion of mobile banking has fundamentally transformed the landscape of financial services, shifting banking activities from physical branches to digital platforms. As mobile banking applications become an integral part of everyday financial behavior, the primary challenge confronted by banks is no longer technology adoption but sustainable long-term usage. Consequently, continuance intention to use mobile banking has emerged as a primary research focus, as it determines the durability of customer relationships and the effectiveness of digital banking investments. In parallel, the accelerated integration of artificial intelligence (AI) into mobile banking applications could advance functionalities such as personalized financial recommendations, intelligent customer service, automated fraud detection, and predictive analytics [1, 2]. These developments suggest that continuance intention is gradually shaped by users' evaluations of AI-driven system capabilities and their trust in intelligent financial technologies.

This concern is particularly salient in rapidly digitalizing economies such as Indonesia, where the usage of mobile banking has continued to be on the rise. Digital Economic Forum 2025 reported that transactions of mobile and digital banking in Indonesia had increased by 52% by 2025, thus reflecting a decisive shift in consumer preferences

toward the channels of digital transaction. According to data from Bank Indonesia, transactions of digital payment reached 3.5 billion and grew 35.3% year-on-year as of January 2025. Mobile banking transactions alone recorded a 29.7% year-on-year increase, while internet banking transactions surged by 19.8% year-on-year. Furthermore, Quick Response Code Indonesia Standard based digital payments flourished by 170.1% year-on-year, supported by a rapid expansion in both users and merchants. From an infrastructural perspective, retail transactions processed through Bank Indonesia Fast Payment increased by 41.5% year-on-year, reaching 338.5 million transactions with a total value of Indonesian Rupiahs 870.9 trillion. In contrast, large-value transactions processed through Bank Indonesia Real-Time Gross Settlement declined by 9% year-on-year [3], indicating a structural shift from traditional interbank transfers toward high-frequency and consumer-driven digital transactions.

These figures underscore not only the widespread adoption of mobile banking in Indonesia but also its transformation into a primary transaction medium for daily financial activities. However, rapid growth in transaction volume does not necessarily imply sustained user commitment. As mobile banking platforms increasingly rely on AI-enabled functionalities, users are required to interact with algorithmic systems that autonomously process sensitive financial data. While artificial intelligence capabilities enhance efficiency and convenience, they may also introduce concerns about data privacy, system reliability, and algorithmic transparency [4–6]. As a result, continuance intention in AI-enabled mobile banking is influenced by a complex interplay between perceived technological competence and users' trust in intelligent systems. Prior research on mobile banking continuance intention chiefly relied on established theories about information systems. While early studies extended the Technology Acceptance Model (TAM) by incorporating satisfaction and trust as post-adoption factors, existing data suggested that perceived usefulness and ease of use positively influenced satisfaction and trust, which subsequently encouraged users' continuance intention toward mobile banking applications [7]. Although these findings offered valuable insights into post-adoption behavior, they were derived primarily within conventional mobile banking contexts and did not explicitly account for the increasing role of AI-driven functionalities in sustaining usage in the long term.

In recent years, considerable concern has arisen over the role of AI in shaping continuance intention. The established body of literature suggested that AI-related characteristics, such as perceived intelligence and anthropomorphism, enhanced users' confirmation and perceived usefulness, which subsequently increased satisfaction and continuance intention [8]. Furthermore, AI characteristics have been shown to strengthen both informational and emotional support, thereby contributing indirectly to users' satisfaction and their willingness to continue using digital services [9]. Additional research has documented that AI-related attributes influenced continuance intention through cognitive and affective mechanisms, although the magnitude and direction of these effects might vary across user groups and generations [10]. Despite these advances, prior studies generally conceived AI through symbolic and perceptual attributes, such as anthropomorphism, perceived intelligence, and human-like interaction [8–10]. Consequently, relatively little attention was paid to examining artificial intelligence capabilities as a comprehensive system-level construct that reflects the functional performance, operational competence, responsiveness, personalization, and learning ability of AI-enabled systems. This limitation points to a broader perspective that could capture how users evaluate the actual capabilities of AI technologies in supporting sustained usage behavior.

Meanwhile, trust has consistently been recognized as a key determinant of mobile banking continuance intention across diverse contexts. Studies conducted in Vietnam and Saudi Arabia demonstrated that trust directly influenced continuance intention and, in some cases, moderated the effects of other post-adoption factors [11–13]. Empirical evidence supports the central role played by trust in shaping users' attitudes and continuance intention across various frameworks of technology acceptance and usage [14]. Nevertheless, existing studies largely positioned trust as a mediating or moderating mechanism, potentially overlooking its direct contribution as a fundamental determinant of continuance intention operating in parallel with technological factors. This approach may underestimate the direct influence of trust, particularly in AI-enabled banking environments characterized by heightened uncertainty and perceived risk. Taken together, these studies revealed several notable research gaps. First, although the application of AI has been examined in mobile banking research, prior studies typically focused on specific AI characteristics rather than conceptualizing artificial intelligence capabilities as holistic system competencies encompassing personalization, learning ability, responsiveness, and automation. Second, existing continuance intention models often depend on complex mediating structures, which might obscure the direct effects of key determinants on continuance intention. Third, despite the robust evidence on the role of trust, few studies examined trust and AI-related technological factors as parallel determinants within a unified post-adoption framework. Moreover, high-growth digital banking markets such as Indonesia, remain underexplored empirically.

To bridge these gaps, the present study investigated artificial intelligence capabilities and trust as direct determinants of continuance intention to use mobile banking. By conceptualizing artificial intelligence capabilities as system-level technological competencies and positioning trust as a core psychological determinant, this study extended post-adoption research beyond traditionally satisfaction- or confirmation-based explanations. This approach was particularly relevant in the Indonesian context, where the rapid expansion of digital transactions underscored

the importance of sustaining user engagement in AI-enabled mobile banking platforms. The novelty of this study lies in three key aspects. First, it shifted the focus from symbolic AI attributes to artificial intelligence capabilities as functional drivers of continuance intention. Second, it treated trust as a primary determinant rather than a secondary mechanism, acknowledging its fundamental role in reducing uncertainty in AI-driven financial services. Third, by providing empirical support from Indonesia, one of the fastest-growing mobile banking markets, the study enriched the predominantly Western- and East Asia-centric literature on mobile banking continuance.

This study contributed theoretically by extending research on continuance intention through the integration of artificial intelligence capabilities and trust within a post-adoption framework. Empirically, it offered quantitative evidence from an emerging digital economy characterized by rapid growth in mobile banking. Practically, the findings provided actionable insights for banks and policymakers by emphasizing that sustained mobile banking usage depended not only on advanced AI functionalities, but also on users' trust in intelligent systems. Aligning technological innovation with trust-building strategies is therefore essential for ensuring the long-term sustainability of AI-enabled mobile banking services.

2 Literature Review and Development of Hypotheses

2.1 Continuance Intention to Use Mobile Banking

Continuance intention refers to an individual's deliberate act to use an information system continuously after initial adoption [15]. In the mobile banking context, continuance intention has gained mounting scholarly attention as a more relevant outcome variable than adoption intention. This shift is particularly evident in mature digital banking environments, where mobile banking services are already widely adopted and integrated into users' daily financial activities. As a result, it is more critical to understand why users persist in using mobile banking applications than understanding the rationale behind their adoption in the first place. The Expectation-Confirmation Model (ECM) provides a widely accepted theoretical foundation for explaining continuance intention [16, 17]. According to ECM, users form continuance intentions based on post-adoption evaluations, which involve the confirmation of prior expectations and the development of favorable beliefs regarding system performance. When users perceive that a system meets or exceeds their expectations, they are more likely to experience positive evaluations that reinforce their intention to use the system persistently. In mobile banking settings, these post-adoption evaluations are no longer limited to basic system functionality but progressively encompass advanced technological features enabled by AI.

As mobile banking applications evolve into AI-enabled platforms, users' continuance intention is incrementally influenced by their perceptions of artificial intelligence capabilities and their confidence in algorithmic processes. AI-driven functionalities, such as automated decision making, personalization, and real-time transaction monitoring, introduce new dimensions to post-adoption evaluations [18–20]. Consequently, users should assess not only whether the system is useful and reliable, but also whether they can trust intelligent technologies to handle sensitive financial information accurately and securely. This transformation underscores the growing importance of both technological competence and psychological assurance in forming continuance intention. In recent years, there has been growing evidence on the importance of digital banking ecosystems, as more mobile banking research emphasized the interplay of continuance intention and long-term user engagement or loyalty. Unlike adoption intention which captured initial acceptance, continuance intention represents repeated usage behavior and users' resistance to alternative platforms. One of the main objectives in contemporary mobile banking research is to identify the technological and psychological factors that drive continuance intention, particularly in the context of rapidly advancing AI-enabled financial services [9, 10].

2.2 Artificial Intelligence Capabilities in Mobile Banking

Artificial intelligence capabilities refer to how users perceive, on the whole, the capacity of an AI-enabled system to perform complex tasks in an effective, accurate, and adaptive manner. In the context of mobile banking, these capabilities are manifested through a range of intelligent functionalities, including personalized financial recommendations, automated and responsive customer service, real-time fraud detection, predictive analytics, and continuous learning from user behavior. Together, these functionalities reflect the extent to which AI systems could enhance service performance, support user decision making, and deliver seamless banking experiences. Unlike isolated AI attributes that emphasize surface-level characteristics, artificial intelligence capabilities could capture users' holistic evaluations of the technological competence and operational effectiveness of AI-enabled mobile banking systems.

Existing research on AI in mobile banking has mainly concentrated on the symbolic or perceptual attributes of AI, such as perceived intelligence and anthropomorphism. For example, studies by Lee et al. [8] and Lin and Lee [9] demonstrated that perceived intelligence and anthropomorphic features influenced users' satisfaction and continuance intention through mechanisms such as expectation confirmation and social support. Similarly, Jisham et al. [10] examined AI characteristics as stimuli that shaped users' cognitive and affective responses within a

stimulus–organism–response framework. This body of research had the undeniable merit of providing valuable insights into how users emotionally and cognitively responded to AI features. Their concurrent emphasis on specific AI characteristics limited an all-inclusive understanding of AI as a functional system, which was capable of delivering sustained value in post-adoption contexts.

In this connection, artificial intelligence capabilities can be conceptualized as a technology-related belief that directly shapes users' evaluations of the effectiveness and usefulness of the system over a long period of time [21, 22]. When users perceive AI-enabled mobile banking systems as capable of delivering accurate, timely, and personalized services, they are more prone to view continued usage as beneficial and efficient. In high-frequency transaction environments, mobile banking is a sector where users rely on applications for routine and time-sensitive financial activities. Therefore, strong artificial intelligence capabilities help reduce cognitive effort and uncertainty. By automating repetitive tasks, anticipating users' needs, and providing intelligent support, AI-enabled systems could enhance users' perceived control and decision quality, which are essential for sustaining long-term engagement [23–25].

Artificial intelligence capabilities also contribute to the perceived reliability and professionalism of mobile banking applications. As users repeatedly interact with intelligent systems, steady performance and adaptive learning reinforce positive evaluations of system competence [9, 26]. These evaluations are particularly salient in financial contexts, where errors or inefficiencies could result in significant perceived risk. Consequently, users exhibit a higher propensity for using mobile banking applications when they believe that the embedded AI systems could support secure, accurate, and efficient transactions. Based on this theoretical reasoning, artificial intelligence capabilities are expected to exert a direct influence on users' continuance intention in mobile banking contexts. Rather than operating solely through indirect mechanisms such as satisfaction or attitude, artificial intelligence capabilities could function as a fundamental technological determinant that shapes users' willingness to maintain long-term usage of AI-enabled mobile banking services. Accordingly, the following hypothesis (**H1**) was proposed:

H1: Artificial intelligence capabilities have a positive effect on continuance intention to use mobile banking.

2.3 Trust in AI-Enabled Mobile Banking

Trust has consistently been identified as a fundamental determinant of user behavior in electronic and mobile banking environments, particularly in contexts characterized by high levels of uncertainty and perceived risk [27–29]. In general, trust reflects users' beliefs that a system is reliable, secure, and capable of acting in their best interest. Within mobile banking, trust traditionally encompasses confidence in the banking institution, its technological infrastructure, and its ability to safeguard sensitive financial information. However, as AI becomes increasingly embedded in mobile banking applications, the scope of trust expands beyond institutional reliability to include trust in intelligent systems that autonomously analyze data, generate recommendations, and execute decisions on behalf of users.

The extant empirical research strongly supports the critical role of trust in determining continuance intention. Bouhlej and Mzoughi [7] reported that trust significantly influenced both user satisfaction and continuance intention toward mobile banking applications, thus highlighting its central position in post-adoption behavior. Similarly, studies conducted in developing economies provided “consistent consensus” that trust directly affected users' intention to continue using mobile banking services. Dang [11] and Nguyen and Dao [12] further revealed that trust not only exerted a direct influence on continuance intention but also moderated the effects of other post-adoption factors, such as user adaptation and perceived usefulness. In addition, Garrouch et al. [13] identified trust as one of the strongest predictors of continuance intention, operating alongside perceived value in mobile banking contexts. Collectively, these findings underscored trust as a robust and stable predictor of sustained mobile banking usage across diverse cultural and economic settings.

Despite its well-established importance, trust is frequently conceptualized as a mediating or moderating variable rather than examined as a primary determinant of continuance intention. This treatment may underestimate the direct impact of trust, particularly in AI-enabled mobile banking environments. AI-driven systems introduce unique challenges, including algorithmic opacity, reduced human intervention, and heightened concerns over data privacy and security [30, 31]. These characteristics amplify users' perceptions of uncertainty and risk, turning trust into a critical psychological mechanism for sustaining long-term engagement. When users lack trust in AI-enabled systems, even highly functional and efficient technologies may fail to retain users over time. From a post-adoption perspective, trust functions as a stabilizing belief that reinforces the user–system relationship. Trust enables users to accept system complexity, rely on automated decision making, and remain confident in the system performance despite limited transparency. Users' willingness to continue using the AI-enabled mobile banking system is crucial to its existence; in this light, trust reduces cognitive and emotional resistance toward intelligent technologies. Accordingly, trust is expected to exert a direct and positive influence on continuance intention, independent of other technological or experiential factors; another hypothesis (**H2**) set out below was proposed:

H2: Trust has a positive effect on continuance intention to use mobile banking.

Synthesizing the afore-mentioned hypotheses, this study proposed a formal research model positing that artificial intelligence capabilities and trust serve as direct determinants of continuance intention in mobile banking applications. This framework reflects a post-adoption perspective and could specifically be tailored to the contemporary banking environment, which is characterized by rapid AI integration and high transaction frequency. The visual representation of this model is shown in Figure 1.

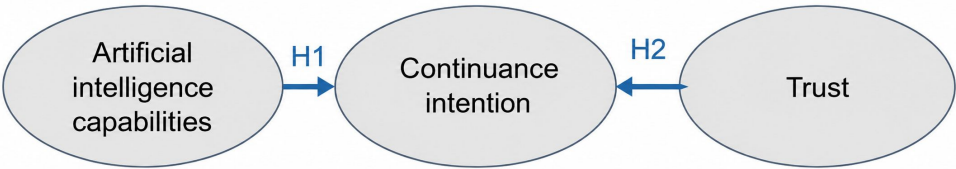


Figure 1. Conceptual framework

3 Methodology

3.1 Research Design and Sample

The research model in Figure 1 reflects a post-adoption perspective, focusing on users’ continued usage behavior rather than initial adoption. This study adopted a quantitative research design with a cross-sectional survey approach to examine the effects of artificial intelligence capabilities and trust on continuance intention to use mobile banking. A quantitative approach is appropriate as the study proposed to test the hypotheses empirically and examine the hypothesized relationships among latent constructs. The setting of this study is Indonesia, one of the fastest-growing digital banking markets in Southeast Asia. Indonesia provides a highly relevant context for examining continuance intention in AI-enabled mobile banking due to the rapid diffusion of digital payment systems, widespread mobile banking usage, and increasing integration of AI in financial services.

The target population consisted of individual users who actively used mobile banking applications and had experience with AI-enabled or AI-supported features, such as automated customer service, personalized financial recommendations, or security-related alerts. To ensure that respondents could provide informed evaluations of continuance behavior, only users who had used mobile banking applications for transactional purposes were included. The sample size was determined using the rule of thumb commonly applied in Structural Equation Modeling (SEM), which recommended a minimum sample size of 10 observations per manifest variable. Given that this study employed 15 manifest variables (measurement items), a minimum of 150 valid responses was required. Accordingly, data were collected from 150 mobile banking users in Indonesia, thus satisfying the recommended threshold for SEM analysis.

3.2 Data Collection Procedure and Measurement of Constructs

The collection of data was carried out by a structured self-administered questionnaire distributed through online channels. Online data collection was chosen due to its suitability for reaching a wide spectrum of mobile banking users and its effectiveness in capturing responses from digitally active individuals. Participation was voluntary, and respondents were informed that their responses would remain anonymous and be used solely for academic research purposes. To minimize common method bias, respondents were assured that there were no right or wrong answers and were encouraged to respond honestly based on their actual experience with mobile banking applications. Incomplete or inconsistent responses were excluded during the data screening process to ensure the partiality and completeness of data.

All constructs in this study were measured using multi-item scales adapted from prior validated studies and modified to fit the context of AI-enabled mobile banking. Items were measured using a five-point Likert scale ranging from 1 (“strongly disagree”) to 5 (“strongly agree”).

The operationalization of the research constructs and measurement items is presented in Appendix Table A1. Artificial intelligence capabilities were measured by items capturing users’ perceptions of AI-driven functionalities in mobile banking, including personalization accuracy, responsiveness, automation, learning ability, and system intelligence. Trust was assessed by items reflecting users’ confidence in the reliability, security, and integrity of AI-enabled mobile banking systems. Continuance intention was evaluated by relevant items assessing users’ intention to continue using mobile banking services in the future.

3.3 Technique of Data Analysis

This study employed Partial Least Squares Structural Equation Modeling (PLS-SEM) for data analysis, using SmartPLS software. PLS-SEM is particularly suitable for this study for several reasons. First, it is appropriate for predictive research models and theory development. Second, PLS-SEM performs well with relatively small to medium sample sizes. Third, it does not require strict assumptions of data normality and is considered suitable for survey-based research. The analysis followed a two-stage approach. First, the measurement model was evaluated by examining indicator reliability, internal consistency reliability, convergent validity, and discriminant validity. Indicator reliability was rated using outer loadings, while internal consistency was evaluated using Cronbach's alpha and composite reliability. Convergent validity was assessed using average variance extracted (AVE), and discriminant validity was examined using the Fornell-Larcker criterion and Heterotrait-Monotrait ratio (HTMT). Second, the structural model was assessed by examining path coefficients, coefficient of determination (R^2), and effect size (f^2). Hypotheses were tested using a bootstrapping procedure with 5,000 resamples to assess the significance of path coefficients.

4 Results

4.1 Profile of the Respondents

A total of 150 valid responses from mobile banking users in Indonesia were included in the analysis. The demographic profile of the respondents was shown in Figure 2. The profile of the respondents demonstrated a diverse and nearly balanced sample, allowing meaningful insights into users' continuance intention toward AI-enabled mobile banking applications.

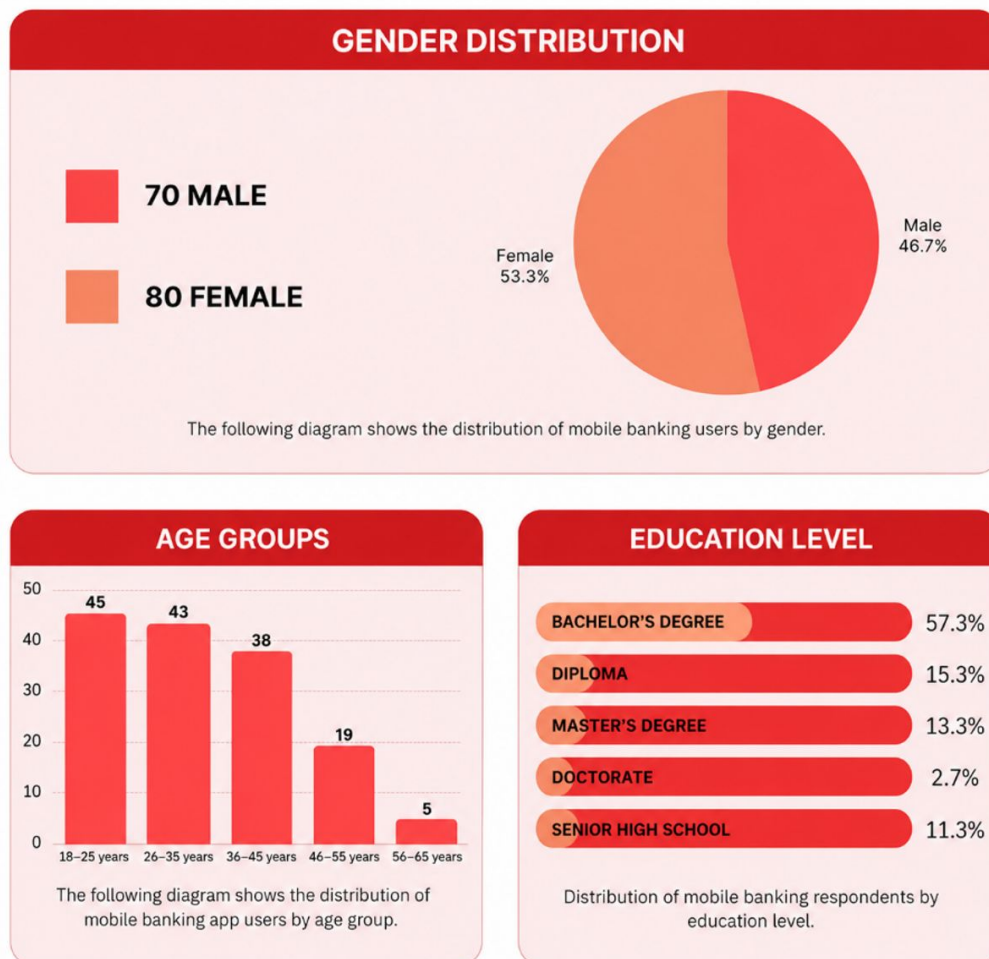


Figure 2. Profile of the respondents

In terms of gender distribution, the sample consisted of 70 male respondents (46.7%) and 80 female respondents (53.3%), indicating a relatively balanced gender composition with a slight predominance of female users. This distribution reflects the broad adoption of mobile banking services across gender groups in Indonesia. Regarding

age, the respondents were mostly within the productive and digitally active age groups. Specifically, 45 respondents (30%) were aged 18–25 years, followed by 43 respondents (28.7%) aged 26–35 years, and 38 respondents (25.3%) aged 36–45 years. Together, these age groups accounted for more than four-fifths of the total sample, suggesting that the majority of respondents were active mobile banking users with substantial exposure to digital financial services. In addition, 19 respondents (12.7%) were aged 46–55 years, while 5 respondents (3.3%) were aged 56–65 years, highlighting that mobile banking usage also extended to older age segments, though to a lesser extent.

In terms of educational background, the sample was dominated by respondents with higher education levels. Most participants held a Bachelor's degree, representing 86 respondents (57.3%), followed by Diploma holders with 23 respondents (15.3%) and Master's degree holders with 20 respondents (13.3%). In contrast, 17 respondents (11.3%) had completed senior secondary education, while only 4 respondents (2.7%) possessed a Doctoral degree. This educational profile suggested that the respondents possessed sufficient cognitive capacity to evaluate AI-enabled mobile banking features and make informed judgments regarding continued usage. Overall, the demographic characteristics of the respondents revealed a heterogeneous yet relevant sample for examining continuance intention in AI-enabled mobile banking contexts. The dominance of digitally active age groups and relatively high educational attainment supports the suitability of the sample for assessing perceptions of artificial intelligence capabilities and trust in mobile banking applications.

4.2 Measurement Model Evaluation

The measurement model was evaluated to assess the reliability and validity of the constructs included in this study. Indicator reliability was examined by analyzing the standardized outer loadings of each measurement item. Based on the initial analysis, items AIC1 and AIC2 from the artificial intelligence capabilities construct, alongside items CI2 and CI3 from the continuance intention to use mobile banking construct, were identified as problematic. Although these indicators exhibited outer loadings above the 0.70 threshold, they demonstrated severe cross-loading issues, where their factor loadings onto their respective parent constructs were lower than their loadings onto other latent variables. The discriminant validity results before item removal are provided in Appendix Tables A2 and A3, which show the initial Fornell–Larcker and HTMT assessments, respectively.

Because this discrepancy critically compromised the subsequent discriminant validity testing, these four items were systematically removed from the model to ensure empirical distinctiveness. As presented in Table 1, after the removal of these items, all remaining indicators exhibited loading values exceeding the recommended threshold.

Table 1. Cross-loadings before and outer loadings after the removal of AIC1, AIC2, CI2, and CI3

Construct	Code	Cross-Loadings Before Item Removal			Outer Loadings After Item Removal
		AIC	CI	TR	
Artificial intelligence capabilities	AIC1	0.884	0.885	0.621	—
	AIC2	0.913	0.926	0.595	—
	AIC3	0.916	0.901	0.563	0.834
	AIC4	0.750	0.600	0.652	0.874
	AIC5	0.748	0.603	0.720	0.880
Continuance intention to use mobile banking	CI1	0.882	0.897	0.637	0.858
	CI2	0.920	0.918	0.610	—
	CI3	0.915	0.914	0.579	—
	CI4	0.674	0.813	0.622	0.905
	CI5	0.681	0.825	0.552	0.910
Trust	TR1	0.705	0.591	0.770	0.765
	TR2	0.572	0.571	0.794	0.789
	TR3	0.517	0.484	0.826	0.828
	TR4	0.543	0.548	0.775	0.783
	TR5	0.525	0.506	0.814	0.815

Note: AIC = artificial intelligence capabilities; CI = continuance intention; TR = trust; — = not calculated because the item was removed.

The retained indicators demonstrated strong psychometric properties, with all standardized loading values exceeding the recommended threshold of 0.7. For the artificial intelligence capabilities construct, the indicator loadings ranged from 0.834 to 0.880, indicating that AI-driven automation, continuous learning, and the overall enhancement of banking experiences were strongly associated with the underlying construct. The trust construct exhibited loading values ranging from 0.765 to 0.828, suggesting that users' perceptions of security, reliability, confidence in automated decision making, and the alignment of the platform with their interests were important dimensions of trust. Similarly, the continuance intention to use mobile banking construct recorded the highest loading values, ranging from 0.858 to 0.910. This revealed that future usage intentions, preference for mobile banking as a primary banking channel, and long-term usage commitment were highly representative of continuance

intention. Simply put, these results confirmed that all retained indicators adequately reflected their respective latent constructs and provided strong evidence of indicator reliability and convergent validity. The suitability of the measurement model was affirmed for subsequent structural model evaluation.

Having established satisfactory indicator reliability through the outer loading assessment, the next step was to evaluate the internal consistency reliability and convergent validity of the constructs. Internal consistency reliability was examined using Cronbach's alpha and composite reliability, while convergent validity was assessed through the AVE. These measures provide further evidence regarding the extent to which the indicators consistently measure their respective constructs and adequately capture the underlying latent variables. The results of the reliability and convergent validity assessment are presented in Table 2.

Table 2. Reliability analysis and AVE results

Latent Variable	Composite Reliability	AVE
Artificial intelligence capabilities	0.897	0.745
Continuance intention	0.921	0.794
Trust	0.896	0.634

Note: AVE = average variance extracted.

The results demonstrated that all constructs possessed a high level of internal consistency reliability, as evidenced by composite reliability values ranging from 0.896 to 0.921, exceeding the recommended threshold of 0.7. A high level of internal consistency among the measurement items was highlighted. Among the constructs, Continuance Intention demonstrated the highest composite reliability value (0.921), followed by artificial intelligence capabilities (0.897) and Trust (0.896). These results confirmed that the indicators consistently measured their respective latent constructs. Furthermore, the AVE values ranged from 0.634 to 0.794, all of which surpassed the recommended minimum threshold of 0.5. Specifically, Continuance Intention exhibited the highest AVE value (0.794), followed by artificial intelligence capabilities (0.745) and Trust (0.634). These findings illustrated that each construct explained more than 50% of the variance in its associated indicators, thereby demonstrating adequate convergent validity. These results support the use of the measurement model as it satisfied the established criteria for reliability and convergent validity. Consequently, the constructs were considered psychometrically sound and suitable for subsequent structural model analysis, as shown in Appendix Table A2.

Following the establishment of indicator reliability, internal consistency reliability, and convergent validity, discriminant validity was further assessed to ensure that each latent construct captured a unique conceptual domain and was empirically distinct from the others. To provide a rigorous evaluation, this study employed two complementary approaches: the traditional Fornell–Larcker criterion and the more stringent HTMT [32]. The initial assessment revealed potential discriminant validity concerns involving two indicators of artificial intelligence capabilities (AIC1 and AIC2) and two indicators of continuance intention (CI2 and CI3). To achieve an acceptable level of discriminant validity, these indicators were removed from the measurement model, and the ultimate results are presented in Table 3.

Table 3. Fornell-Larcker and Heterotrait-Monotrait ratio assessment results

Construct	Artificial Intelligence Capabilities	Continuance Intention	Trust
Artificial intelligence capabilities	0.863*	0.869	0.875
Continuance intention	0.757	0.891*	0.780
Trust	0.738	0.681	0.796*

Note: Diagonals (in bold with *) represent the square root of average variance extracted; lower triangle values represent latent construct correlations; upper triangle values represent Heterotrait-Monotrait ratio (HTMT) values.

The discriminant validity assessment provided evidence that the latent constructs were empirically distinguishable. According to the Fornell-Larcker criterion, discriminant validity was established when the square root of the AVE for each construct exceeded its highest correlation with any other construct. The diagonal elements showed that the square roots of AVE for artificial intelligence capabilities (0.863), continuance intention (0.891), and trust (0.796) were consistently greater than the corresponding inter-construct correlations, thereby satisfying the Fornell-Larcker criterion. To complement this assessment, the HTMT ratio was examined. The highest HTMT value was observed between artificial intelligence capabilities and trust (0.875), followed by artificial intelligence capabilities and continuance intention (0.869), while the relationship between trust and continuance intention yielded a value of 0.780. Although the first two HTMT values slightly exceeded the conservative threshold of 0.85, they remained below the more liberal threshold of 0.9 recommended for conceptually related constructs [32].

Given the theoretical context of AI-enabled mobile banking, such associations were expected because users' perceptions of artificial intelligence capabilities including personalization, responsiveness, and service automation were closely linked to the development of trust, which in turn influenced their intention to continue using the service. Therefore, the observed level of construct overlap reflects theoretically meaningful relationships rather than a lack of discriminant validity. Overall, the results from both the Fornell-Larcker and HTMT assessments confirmed that the latent constructs exhibited adequate discriminant validity and could be considered empirically distinct.

4.3 Structural Model Evaluation

The structural model was evaluated to examine the proposed hypotheses and assess the model's explanatory power. As reported in Appendix Table A4, the R^2 for continuance intention is 0.606, with an adjusted R^2 value of 0.601. According to established PLS-SEM guidelines, this result validated a moderate level of explanatory power, suggesting that artificial intelligence capabilities and trust jointly explained 60.6% of the variance in users' continuance intention to use mobile banking. This finding demonstrated that the proposed model provided substantial explanatory capacity, despite its relatively parsimonious structure. To further assess the contribution of each predictor, f^2 values were examined to assess the effect size of each predictor (Appendix Table A5). The results revealed that artificial intelligence capabilities exerted a large effect on continuance intention ($f^2 = 0.362$). This construct was shown to contribute substantially to explaining users' intentions to use mobile banking services continuously. In contrast, Trust exhibited a small effect on continuance intention ($f^2 = 0.083$), suggesting that although trust remained an important predictor, its incremental contribution to the explained variance was comparatively small. To conclude, these findings verified the adequacy of the structural model and thus supported the dominant role of artificial intelligence capabilities in affecting continuance intention.

After establishing the adequacy of the measurement model, the structural model was employed to test the proposed hypotheses and examine the hypothesized relationships among the latent constructs. The significance of the structural paths was assessed using a bootstrapping procedure with 5,000 resamples. Path coefficients were evaluated based on their corresponding t-statistics and p-values, with statistical significance determined at the 5% significance level. The results of the hypothesis testing are presented in Figure 3 and Table 4.

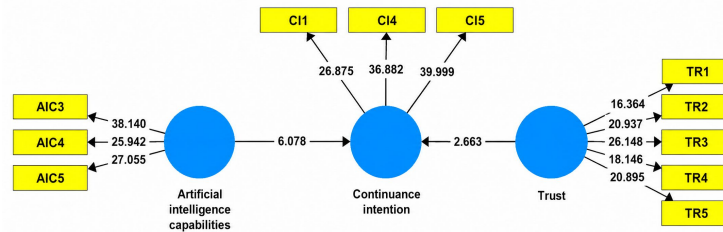


Figure 3. Structural model of artificial intelligence capabilities, continuance intention and trust

Note: The values on the arrows represent t-statistics

Table 4. Testing of hypotheses

Hypothesis	Original Sample	Sample Mean	Standard Deviation	t-Statistics	p-Values
Artificial intelligence capabilities → continuance intention	0.560	0.574	0.092	6.078	0.000
Trust → continuance intention	0.267	0.257	0.100	2.663	0.008

The results revealed that both proposed hypotheses were supported. Artificial intelligence capabilities exerted a significantly positive influence on continuance intention ($t = 6.078$, $p = 0.000$), suggesting that users who perceived mobile banking applications as possessing advanced AI-driven features, such as personalization, responsiveness, and service automation, were more likely to maintain their usage over time (as Appendix Table A5). This finding shed light on the importance of technological capabilities in enhancing users' perceptions of value and convenience, thereby encouraging continued engagement with mobile banking services. On the other hand, Trust also demonstrated a significantly positive effect on continuance intention ($t = 2.663$, $p = 0.008$), indicating that users' confidence in the reliability, security, and dependability of AI-enabled mobile banking services all played a crucial role in fostering long-term usage intentions (as Appendix Table A5). A comparison of the t-statistics suggested that artificial intelligence capabilities exerted a stronger influence on continuance intention than trust, although both constructs

contributed significantly to explaining user behavior. Overall, these findings provided empirical evidence to prove that both technological and psychological factors were essential determinants of sustained mobile banking usage.

5 Discussion

This study examined the roles of artificial intelligence capabilities and trust in shaping users' continuance intention toward mobile banking services. The findings provided strong empirical evidence that both technological competence and psychological assurance were critical determinants of post-adoption behavior in AI-enabled financial services. The significant effects of AI and Trust suggested that sustaining mobile banking usage extended beyond traditional post-adoption factors, such as perceived usefulness and satisfaction, and increasingly depended on users' evaluations of the intelligent system performance and their confidence in AI-driven service delivery. The positive relationship between AI and CI demonstrated that users placed substantial importance on the operational effectiveness of AI technologies embedded within mobile banking applications. Specifically, users appeared to evaluate AI not merely as an innovative technological feature, but also as a functional capability that enhances service quality through personalization, responsiveness, automation, and continuous learning. This finding complemented prior research emphasizing AI characteristics such as perceived intelligence, anthropomorphism, and human-like interaction [8, 9]. While earlier studies primarily focused on the emotional and social dimensions of AI adoption, the present study reached the conclusion that long-term usage decisions were more strongly influenced by users' assessments of how effectively AI performed core banking functions.

This finding is particularly relevant in the context of mobile banking, where service interactions occur frequently and often involve financially sensitive transactions. Unlike other digital services that may be used occasionally, mobile banking requires continuous user engagement for payments, fund transfers, account monitoring, and financial planning. In such environments, artificial intelligence capabilities contribute directly to reducing cognitive burden, minimizing operational errors, accelerating transaction processes, and providing contextually relevant recommendations. Consequently, users perceive greater efficiency, convenience, and control, all of which strengthen their intention to continue using the service. These findings support the proposition that artificial intelligence capabilities represented a distinct technology-related belief that directly influenced continuance behavior, rather than operating solely through mediating constructs such as satisfaction or attitude. The results also provided a possible explanation for inconsistencies observed in prior technology research. Traditional models such as TAM and ECM generally assume that perceived usefulness and satisfaction serve as the primary drivers of post-adoption behavior. However, empirical findings have not always been consistent in AI-enabled environments. Previous research reported that perceived usefulness did not always successfully predict continuance intention in digital banking settings [10]. The present findings further suggest that as intelligent technologies became increasingly embedded within service delivery systems, users might evaluate technological competence more directly. Consequently, artificial intelligence capabilities emerged as a more appropriate explanatory factor than traditional utility-based beliefs because they capture the actual performance attributes that users experience during interactions in service delivery.

Apart from technological competence, trust exerted a significantly positive effect on continuance intention. This finding is consistent with previous studies that identified trust as a key determinant of mobile banking adoption and continuance behavior [7, 13, 14]. However, this study has shifted the focus of existing literature by conceptualizing trust as a direct antecedent of continuance intention rather than merely a mediating or moderating variable. This distinction is theoretically important because it positions trust as a fundamental belief that directly stabilizes user relationships with AI-enabled financial systems. The significance of trust becomes increasingly evident as mobile banking platforms rely heavily on automated decision-making and data-driven processes. AI-enabled systems frequently collect, analyze, and utilize large volumes of personal and financial information to deliver personalized services. While these capabilities enhance efficiency and customer experience, they simultaneously increase concerns regarding privacy, security, transparency, and algorithmic accountability. Users are often unable to fully understand how algorithmic decisions are generated, thus creating conditions of uncertainty commonly described as the "black-box" problem of AI. Under such circumstances, trust functions as a risk-reduction mechanism that enables users to accept vulnerability and continue relying on the system, despite having incomplete knowledge of its internal processes.

The findings are in conformity with evidence obtained from emerging economies, where trust has been constantly identified as a critical determinant of using digital financial services. Previous studies advocated that trust contributed significantly to continuance intention by reducing perceived uncertainty and strengthening users' confidence in technology-mediated transactions [11, 12]. The present study corroborated this evidence and further demonstrated that trust remained largely relevant, even in the presence of sophisticated artificial intelligence capabilities. Therefore, technological advancement alone may not be sufficient to ensure sustained usage if users continue to harbor concerns regarding the security, reliability, and fairness of the system. An interesting insight emerging from this study is the complementary role of artificial intelligence capabilities and Trust in explaining continuance intention. Although artificial intelligence capabilities exerted a stronger statistical effect, both constructs significantly contributed to

users' intentions to continue using mobile banking services. This finding suggested that continuance intention was not driven by a single dominant factor but rather by the combined influence of technological performance and relational assurance. From a theoretical perspective, this supported a more holistic understanding of post-adoption behavior in AI-enabled environments. The findings challenged the assumption that technological superiority alone was sufficient to sustain user engagement. Even highly advanced AI systems may struggle to retain users if they are perceived as opaque, unreliable, or risky. Conversely, trust in the absence of adequate technological performance may also be insufficient, as users ultimately expect to receive tangible functional benefits from digital financial services. This dual requirement reflects the evolving nature of technology continuance in the era of AI, when users simultaneously evaluate both system competence and trustworthiness when making decisions regarding continued usage. Consequently, sustained engagement with AI-enabled mobile banking services depends on achieving a balance between technological excellence and users' confidence in the integrity and reliability of the service provider.

To conclude, this study extended current continuance intention theories, particularly the ECM and technology continuance frameworks. Traditional continuance models primarily emphasize cognitive evaluations associated with confirmation, usefulness, satisfaction, and perceived performance. However, the present findings proved that post-adoption beliefs in AI-driven contexts were multidimensional and included evaluations of intelligent technological capabilities as well as trust in algorithmic systems. By integrating these dimensions, this study contributed to the growing body of literature seeking to explain user behavior in increasingly autonomous and data-intensive service environments. From a managerial perspective, several practical implications emerge. First, financial institutions should prioritize investments in artificial intelligence capabilities that generate observable value for users. Features such as intelligent personalization, predictive financial assistance, automated customer service, fraud detection, and adaptive recommendation systems are likely to enhance users' perceptions of technological competence and encourage continued usage. Merely incorporating AI as a symbolic innovation without delivering meaningful functional improvements may have limited impact on user retention. Second, banks should actively strengthen trust-building mechanisms alongside technological innovation. Given the increasing complexity of AI-driven services, organizations should emphasize transparency, explainability, security, and ethical AI governance. Clear explanations of how customer data are collected, processed, and protected could reduce uncertainty and strengthen user confidence. Furthermore, implementing robust cybersecurity measures and maintaining responsive customer support systems could reinforce perceptions of reliability and institutional credibility. Finally, these implications are particularly relevant in rapidly digitalizing markets such as Indonesia, where mobile banking adoption continues to expand at an unprecedented rate. As competition among banks and fintech providers intensifies, retaining existing users becomes as urgent as acquiring new customers. The findings suggested that sustainable competitive advantage requires organizations to simultaneously enhance artificial intelligence capabilities and cultivate trust among customers. By integrating technological excellence with relational assurance, financial institutions could foster stronger continuance intention and ensure long-term sustainability of AI-enabled mobile banking services.

6 Conclusions

This paper examined the determinants of continuance intention in the use of mobile banking services. The roles of artificial intelligence capabilities and trust in an AI-enabled banking context were underscored. Drawing on a post-adoption perspective and employing a SEM approach, this study presented empirical evidence from Indonesia, one of the fastest-growing digital banking markets. The findings demonstrated that both artificial intelligence capabilities and trust significantly influenced users' continuance intention; this confirmed that sustained usage of mobile banking services was shaped by a combination of technological competence and psychological assurance. The results indicated that users evaluated AI-enabled mobile banking not merely based on the presence of intelligent features, but on the functional effectiveness of artificial intelligence capabilities, such as personalization accuracy, responsiveness, and automation. These practical capabilities reduced cognitive effort, enhanced service efficiency, and improved the overall user experience. Therefore, users were more willing to continue using mobile banking applications. The current findings extended prior research by emphasizing that holistic AI capabilities, rather than merely symbolic AI characteristics, function as a direct driver of continuance intention.

While research on the occurrence of this interrelation was at a preliminary stage, the study validated trust as a core determinant of post-adoption behavior. Trust directly influences continuance intention, thus highlighting its foundational role in AI-enabled mobile banking environments characterized by perceived uncertainty, algorithmic opacity, and heightened privacy concerns. By positioning trust as a primary predictor rather than a mediating or moderating variable, this study advanced the theoretical understanding of continuance intention and focused on the importance of psychological assurance in sustaining long-term user engagement. Finally, this study contributed to the literature on mobile banking and information systems by integrating AI capability and trust within a parsimonious post-adoption framework. Practical implications were offered for banks and fintech providers, i.e., investments in advanced AI technologies should be complemented by trust-building strategies to ensure the long-term sustainability of mobile banking services, particularly in rapidly digitalizing markets like Indonesia.

7 Limitations and Future Research Directions

Despite its contributions, this study has several limitations that should be resolved in future. First, the study employed a cross-sectional research design, which could only partially capture changes in users' perceptions and behavior over time. Continuance intention is inherently dynamic, and future studies may adopt longitudinal designs to examine how artificial intelligence capabilities and trust evolve as users gain more experience with AI-enabled mobile banking systems. Second, the data were collected using a self-reported survey, which may be subject to common method bias and social desirability effects. Although procedural remedies were implemented, future research could incorporate objective usage data or adopt mixed method approaches to deepen the understanding of continuance behavior. Third, this study used a parsimonious model consisting of two primary determinants of continuance intention. Although this approach enhanced theoretical clarity, it might not wholly capture the complexity of post-adoption behavior.

Future studies could be incorporated with additional constructs, such as user satisfaction, perceived risk, or AI transparency, to further enrich the explanatory power of continuance intention. Fourth, the empirical context of this study was confined to Indonesia. While Indonesia represents a highly relevant and dynamic digital banking market, differences observed in the cultural, regulatory, and technological aspects may influence users' perceptions of AI and trust.

Cross-country or comparative studies could be conducted to extend the generalizability of the findings across various economic and cultural settings. Finally, this study treated artificial intelligence capabilities as a unified construct. Despite the encouraging results in this early attempt, future research may disaggregate artificial intelligence capabilities into more specific dimensions to examine their differential effects on trust and continuance intention. Such investigations could provide more granular insights into the aspects of AI which are most critical for sustaining long-term usage in mobile banking.

Author Contributions

Conceptualization, N.S.W. and S.S.S.; methodology, T.H.; software, S.S.S.; validation, G.C.P. and T.H.; formal analysis, S.S.S.; investigation, N.S.W.; resources, N.S.W.; data curation, S.S.S.; writing—original draft preparation, N.S.W. and S.S.S.; writing—review and editing, T.H. and G.C.P.; visualization, S.S.S.; supervision, G.C.P.; project administration, N.S.W. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used generative AI and AI-assisted technologies solely to improve the grammar, phrasing, and overall readability of the text. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the framework, integrity, and accuracy of the final publication.

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Appendix

Table A1. Operationalization of research constructs and measurement items

Construct	Code	Measurement Items
Artificial intelligence capabilities	AIC1	The mobile banking application uses AI to provide personalized services tailored to my needs.
	AIC2	The AI features in the mobile banking application respond quickly and accurately to my requests.
	AIC3	The mobile banking application effectively uses AI to automate financial services.
	AIC4	The AI system in the mobile banking application continuously learns to improve service quality.
	AIC5	Overall, the artificial intelligence capabilities of the mobile banking application enhance my banking experience.
Trust	TR1	I trust the mobile banking application to handle my financial transactions securely.
	TR2	I believe the AI-enabled mobile banking system is reliable.
	TR3	I feel confident in using the mobile banking application even when decisions are automated.
	TR4	The mobile banking application acts in my best interest.
	TR5	I trust the AI features of the mobile banking application.
Continuance intention to use mobile banking	CI1	I intend to continue using the mobile banking application in the future.
	CI2	I will use mobile banking application frequently in the future.
	CI3	I will not switch to alternative banking channels if mobile banking remains available.
	CI4	I expect to keep using this mobile banking application as my primary banking channel.
	CI5	My use of this mobile banking application will continue in the long term.

Table A2. Fornell-Larcker criterion before the removal of AIC1, AIC2, CI2, and CI3

Construct	Artificial Intelligence Capabilities	Continuance Intention	Trust
Artificial intelligence capabilities	0.846	—	—
Continuance intention	0.941	0.875	—
Trust	0.726	0.683	0.796

Note: — indicates not applicable.

Table A3. Heterotrait-Monotrait ratio (HTMT) before the removal of AIC1, AIC2, CI2, and CI3

Construct	Artificial Intelligence Capabilities	Continuance Intention	Trust
Artificial intelligence capabilities	—	—	—
Continuance intention	0.999	—	—
Trust	0.841	0.766	—

Note: — indicates not applicable.

Table A4. Coefficient of determination (R^2) results

Construct	R^2	R^2 Adjust
Continuance intention	0.606	0.601

Table A5. Effect size (f^2) results

Structural Path	f^2 Value	Effect Size
Artificial intelligence capabilities → continuance intention	0.362	Large
Trust → continuance intention	0.083	Smaill